PWM-Driven RL Circuit Modelling

Let

$$I_0 = \frac{V_s}{R}, \qquad \tau = \frac{L}{R}, \qquad T = \frac{1}{F}, \qquad D = \frac{T_{ON}}{T}.$$

1. ON Interval

The governing equation is

$$L\frac{di}{dt} + Ri = V_s.$$

Its solution is

$$i(t) = I_0 + (I_{\min} - I_0)\, e^{-t/\tau}, \qquad 0 \le t \le DT.$$

Hence

$$I_{\max} = I_0 + (I_{\min} - I_0)\, e^{-DT/\tau}.$$

2. OFF Interval

The governing equation is

$$L\frac{di}{dt} + Ri = 0.$$

whose solution is

$$i(t) = I_{\max} e^{-t/\tau}, \qquad 0 \le t \le (1-D)T.$$

Thus

$$I_{\min} = I_{\max} e^{-(1-D)T/\tau}.$$

3. Derivation of $I_{\max}$ and $I_{\min}$

Substitute the OFF-state expression into the ON-state boundary equation:

$$I_{\max} = I_0 + \Big(I_{\max} e^{-(1-D)T/\tau} - I_0\Big)\, e^{-DT/\tau}.$$

Using

$$e^{-DT/\tau} e^{-(1-D)T/\tau} = e^{-T/\tau},$$

gives

$$I_{\max}(1 - e^{-T/\tau}) = I_0(1 - e^{-DT/\tau}),$$

hence

$$\boxed{I_{\max} = I_0 \frac{1 - e^{-DT/\tau}}{1 - e^{-T/\tau}}}$$

and

$$\boxed{I_{\min} = I_0 \frac{e^{-(1-D)T/\tau}(1 - e^{-DT/\tau})}{1 - e^{-T/\tau}}}$$

4. Piecewise Current

$$i(t) = \begin{cases} I_0 + (I_{\min} - I_0)e^{-t/\tau}, & 0 \le t < DT, \\ I_{\max}e^{-(t-DT)/\tau}, & DT \le t < T. \end{cases}$$

5. Exact Average Current

$$I_{\text{avg}} = \frac{1}{T}\left[\int_0^{DT} i_{\text{ON}}(t)\,dt + \int_{DT}^{T} i_{\text{OFF}}(t)\,dt\right].$$

The ON integral is

$$I_0 DT + \tau(I_{\min} - I_0)(1 - e^{-DT/\tau}).$$

The OFF integral is

$$\tau I_{\max}(1 - e^{-(1-D)T/\tau}).$$

Substituting $I_{\max}$ and $I_{\min}$,

$$\boxed{I_{\text{avg}} = \frac{V_s}{R}\left[D - \frac{\tau}{T}\frac{(1 - e^{-DT/\tau})(1 - e^{-(1-D)T/\tau})}{1 - e^{-T/\tau}}\right].}$$

6. Ripple

$$\Delta I = I_{\max} - I_{\min}$$

which becomes

$$\boxed{\Delta I = \frac{V_s}{R}\frac{(1 - e^{-DT/\tau})(1 - e^{-(1-D)T/\tau})}{1 - e^{-T/\tau}}}$$

or

$$\Delta I = \frac{V_s}{R}\frac{(1 - e^{-D/(F\tau)})(1 - e^{-(1-D)/(F\tau)})}{1 - e^{-1/(F\tau)}}.$$

For $F\tau \gg 1$,

$$\boxed{\Delta I \approx \frac{V_s}{LF}D(1 - D).}$$

Since

$$\frac{d}{dD}[D(1 - D)] = 1 - 2D,$$

maximum ripple occurs at

$$\boxed{D = 0.5.}$$

7. Error Metrics

$$E_{\text{ripple}} = \pm\frac{\Delta I}{2},$$

$$E_{\text{up}} = I_{\max} - I_{\text{target}},$$

$$E_{\text{down}} = I_{\text{target}} - I_{\min},$$

$$E_{\text{worst}} = \max(E_{\text{up}}, E_{\text{down}}).$$